Paranidharan Muruganantham

AI Research Engineer | End-to-End Foundation Model Training | TPU & GPU at Scale | Bengaluru, India
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Professional Summary

Owns foundation-model training end to end, from synthetic data generation through pretraining to post-training. Core member of the team behind **Dhīr 2B and Dhīr 5B MoE**, trained from scratch on 6T+ tokens and launched by India's Ministry of Electronics and IT under the IndiaAI Mission in November 2026. Runs the full training stack on a **2,048-chip Cloud TPU v6e** cluster, from XPK provisioning and Lustre/GCS storage to JAX/MaxText training and XProf profiling, in close collaboration with Google Cloud engineers. Also trains on **multi-GPU** systems (PyTorch, DeepSpeed, FSDP, Megatron-LM). Previously at IISc, on research with IBM and DRDO.

Experience

ZenteiQ AiTech Innovations Pvt. Ltd. — Bengaluru, India
AI Research Engineer *(owns model training and data generation)* — Aug 2025 – Present

- Made the core pretraining decisions (**architecture, data mixture and compute allocation**) for **Dhīr 2B** (dense) and **Dhīr 5B MoE** (5.09B total, 1.79B active, top-2 of 12 experts), trained from random initialisation on **6T+ tokens** and released open-source on AIKosh.
- Own model training and every pretraining run on a **2,048-chip Google Cloud TPU v6e** cluster (JAX, MaxText, XLA).
- Run the training infrastructure end to end: provision TPU slices on GKE with **XPK** and **Cluster Toolkit**, stage data and checkpoints on **Managed Lustre** and **Cloud Storage (GCS)**, and profile runs with **XProf**.
- Work directly with **Google Cloud engineers** to diagnose and resolve TPU cluster, networking and runtime issues.
- Reached **41% peak MFU** in throughput ablations; ran production pretraining at **38% MFU**.
- Trained a **9B-parameter** model through **20 days of continuous pretraining**, the team's next scale-up beyond Dhīr.
- Designed and ran the pretraining ablations on architecture and data mixture that set the final-run configuration.
- Own data generation through **DSDG**, a distributed pipeline in which multiple LLMs expand seed data into synthetic corpora for English and 10 Indian languages; set the data team's specifications (context length, format, coverage).
- Contributed to post-training (SFT, RLHF, GRPO, DPO); models average **61.4** (2B) and **64.6** (MoE) over 41 benchmarks.

Indian Institute of Science (IISc), AiREX Lab — Bengaluru, India
ML Research Associate *(Advisor: Prof. Sashikumaar Ganesan)* — Dec 2024 – Aug 2025

- Worked with **IBM** on **Granite** foundation models in a research collaboration, focusing on transformer architectures.
- Built a radar-signal classifier for **DRDO** that improved performance by **35%**.
- Built its data pipeline on **Kafka** and containerised the workflow with Docker (CommitLint, Black).
- Wrote the regression module, PyPI packaging and documentation for SciREX, an open-source scientific ML library.

Indian Institute of Science (IISc), AiREX Lab — Bengaluru, India
Research Intern — Jun 2024 – Aug 2024

- Built a multimodal (vision-language) RAG system for information retrieval and response generation; Celery and Redis workers cut processing time by **40%**.
- Scraped and processed **500K+** records and **50 GB+** of raw data with Scrapy for training and inference.

Dhīr Release & Write-ups

- **Release:** Dhīr model page | AIKosh (national AI repository) | MeitY launch coverage, The Tribune | Launch feature
- **ZenteiQ research blog:** How we trained Dhīr | Reinforcement learning on TPUs | The price of a token

Selected Work

Phyen: 8.3B LLM for physics and engineering reasoning — Full fine-tuning | Hugging Face

- Fully fine-tuned a Qwen-based model end to end on physics and engineering reasoning (thermodynamics, fluid mechanics, structural analysis), updating all weights rather than training an adapter.
- Merged and released the result as a standalone BF16 checkpoint (sharded safetensors) that loads directly with Hugging Face Transformers, with no base model or adapter needed.
- Kept the base model's general language ability while prioritising scientific precision.

Model optimisation: PyTorch → ONNX → TensorRT — CUDA, PyCUDA, TensorRT

- Optimised YOLOv5 inference on an NVIDIA RTX A6000 with TensorRT FP16: **1.9**× faster than PyTorch and **5.2**× faster than ONNX Runtime (495 vs 95 FPS).
- Cut latency by **46%** (3.76 ms to 2.02 ms) through CUDA memory management and stream synchronisation.

Physics VLM: Qwen2-VL-7B for visual physics problem solving QLoRA | DeepSpeed ZeRO-3

- Fine-tuned Qwen2-VL-7B-Instruct on ScienceQA to read physics problems from images and answer in structured JSON.
- Used a multi-adapter design: froze the vision encoder to preserve its pretrained representations, trained the language model with 4-bit QLoRA, and gave the vision-language projector a small LoRA adapter (rank 4–8).
- Trained with DeepSpeed ZeRO-3, tracked runs in Weights & Biases, and released the adapters under Apache 2.0.

Document Prep Kit: LLM training data from scientific papers IBM Docling, Python

- Built a data-preparation toolkit that turns scientific papers into fine-tuning data for LLMs, keeping the structure of figures, tables, formulas and captions intact instead of flattening them to plain text.
- Used IBM Docling layout parsing and proximity-based caption extraction to tie figures and tables to their text.
- Generated additional context with a language model to drive question-answer pair generation from each document, improving the pipeline's yield of usable training examples.
- Designed as modular, object-oriented components with structured logging, so new document types and extraction steps can be added and debugged independently.

Open Source

SciREX: Scientific Machine Learning library (AiREX Lab, IISc) Python, PyPI

- Developed the library's complete regression module within its machine-learning section.
- Added PyPI installation support and wrote the full user and developer documentation.

Technical Skills

Pretraining & scaling	JAX, MaxText, XLA, Google Cloud TPU v6e, mixture-of-experts, MFU optimisation, scaling ablations, data mixtures
Training infrastructure	XPK, Cluster Toolkit, GKE, Managed Lustre, Cloud Storage (GCS), XProf profiler, Docker, Weights & Biases
Distributed training	PyTorch, Megatron-LM, DeepSpeed (ZeRO), FSDP, DDP, Ray; BF16/FP16 mixed precision, activation recomputation
Post-training	SFT, RLHF, GRPO, DPO, LoRA/QLoRA, PEFT, Hugging Face Transformers
Data	Synthetic data generation, datatrove, NeMo Curator, IBM Docling, Scrapy, Kafka
Optimisation & inference	NVIDIA GPUs, CUDA, PyCUDA, TensorRT, ONNX
ML & tooling	TensorFlow, scikit-learn, NumPy, pandas, FastAPI, Celery, Redis, LangChain
Languages & cloud	Python, C++, Google Cloud, AWS (EC2, S3, Lambda, EKS), Docker, GitHub Actions, Git

Community

- **Lead organiser,** CASML 2025 (IISc): international scientific machine-learning conference with 500+ participants, including a hands-on LLM and JAX workshop on Google TPUs.
- **Lead organiser,** CASML 2024 (IISc): first international scientific ML conference in India, 300+ participants.
- **Exhibitor for ZenteiQ,** India AI Impact Summit 2026 (New Delhi): presented ZenteiQ's foundation-model work to founders, researchers and industry leaders.

Education

Sathyabama Institute of Science and Technology Chennai, India
B.E. Computer Science and Engineering, First Class with Distinction 2020 – 2024